



Letter

Lyapunov landscape elucidates disorder-taming-chaos in arrays of coupled pendula

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ABSTRACT

Disordering coupled arrays of nonlinear elements can convert chaotic dynamics to periodic dynamics. Disorder-taming-chaos (DTC) was originally demonstrated by disordering the lengths of torqued and damped coupled pendula. Guided by a possibility space of the lengths' mean and standard deviation, we investigate periodicity and chaos in such arrays by estimating their maximal Lyapunov exponents via numerical integration of the corresponding variational equations, after identifying a plateau in the variational algorithm's parameter space where these exponents are well defined. In mapping the Lyapunov landscape as a function of disorder and coupling, we classify the dynamics as periodic or chaotic—exhaustively for binary and trinary arrays and typically for longer arrays, including the continuum limit—and connect with the arrays' spatiotemporal patterns. We thereby locate DTC at bifurcations of the single-pendulum dynamics and find that both mean-changing and mean-preserving length-disorders can tame chaos, but the latter may require large disorder at large coupling.

1. Introduction

The pendulum is a canonical nonlinear system at the core of physics [1–3] that continues to generate interest [4–7]. Coupled pendula [8–10] and their quantum and superconducting analogues [11–13] are phenomenologically rich. They are laboratories for studying diverse physics, including periodic and chaotic dynamics.

In 1995, Braiman et al. [14] demonstrated how disordering the lengths of pendula in a torqued, damped, coupled array can convert chaotic dynamics into periodic dynamics. Chaos plus disorder equals periodicity is a remarkable equation. Subsequently, Lindner et al. [15] systematically and computationally explored disorder-taming-chaos (DTC) and discovered how coupling and torque can minimize the array's maximal Lyapunov exponent, a measure of its chaoticity. The existence of an optimal disorder that minimizes the largest Lyapunov exponent of the array is analogous to stochastic resonance [16,17], where an optimal noise can maximize a signal-to-noise ratio, which has applications from ice ages [18] to hair cells [19]. Both phenomena involve optimal behavior (suitably-defined) for a nonzero value of a stochastic influence (disorder or noise).

In more recent work, Gavrielides et al. [20] showed that under certain conditions even a single impurity in a pendulum array can convert complex chaotic behavior to simple spatiotemporal patterns. Wiess

et al. [21] used single impurities to synchronize two-dimensional arrays even when all the pendula *and the impurity* were chaotic when uncoupled. Qi et al. [22] used a correlation function as an order parameter to describe the effect of disorder on the pendulum array. Shew et al. [23] realized DTC experimentally with mechanical pendula. Meanwhile, C et al. [24] explored the effect of disorder and chaos on quantum many-body systems, while Barioni et al. [25] described disorder-promoted synchronization in coupled laser networks.

In this article, we revisit DTC in pendulum arrays with modern high-performance computing and extend previous results through full ranges of coupling and disorder, for a variety of *types* of disorder, including very short pendula where rapid end-over-end motion is numerically difficult to follow. We demonstrate the robustness of a variational algorithm for estimating maximal Lyapunov exponents and find islands of order in seas of chaos. We extend the system to the infinite coupling limit where the arrays homogenize, and we generalize the equations of motion to the continuum limit of infinite array lengths. We present the Lyapunov landscape, which reveals the robustness of DTC even for mean-preserving disorders, where small coupling promotes DTC, but large coupling impedes it.

Section 2 locates DTC at bifurcations of the single-pendulum dynamics. Section 3 introduces our equations of motion and types of disorder and organizes the disorder by the infinite coupling limit, where adjacent

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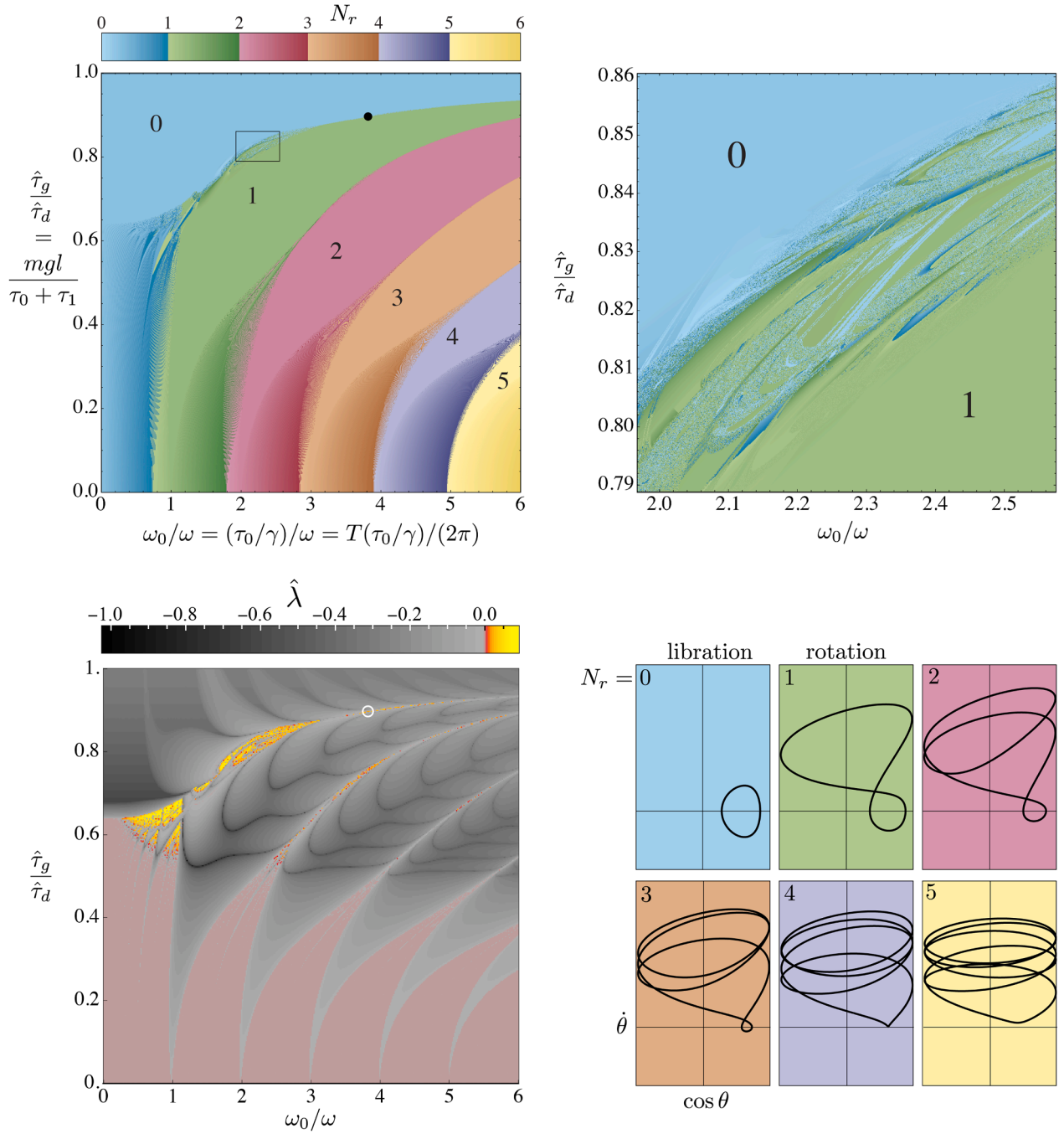


Fig. 1. Top row: Pendulum dynamics in a length l versus torquing period $T = 2\pi/\omega$ parameter space slice, nondimensionalized with the other system parameters. Bands correspond to periodic motion, of N_r mean rotations per torquing period, separated by chaotic bifurcations. Gradient band palette indicates that rotations N_r increase continuously with period T for short lengths l (bottom edge) but decrease discontinuously with increasing length l for long periods (right edge). Black dot indicates default parameters straddling libration and rotation. Top right: Closeup of rectangle region suggests fractal structure. Bottom right: Corresponding $\{\cos \theta, \dot{\theta}\}$ compactified phase space slice has flows with an extra loop for each successive band. Bottom left: Magma palette indicates corresponding maximal Lyapunov exponent estimates $\hat{\lambda}$. Positive exponents (reds and yellows) between periodic bands (grays) indicate chaos. Dynamics are exactly solvable in the zero-length limit (bottom black edge). White circle marks default parameters.

pendula are fused together. Section 4 introduces the variational algorithm to estimate the arrays' maximal Lyapunov exponents. Section 5 classifies dynamics as periodic or chaotic as a function of disorder and coupling (for short, generic, and infinite arrays), connects with the arrays' spatiotemporal patterns, and demonstrates DTC in the special case of mean-preserving disorders. Section 6 offers some conclusions, including a possible application. Appendices summarize our notation, detail

our numerical methods, and verify the “easy” DTC of mean-changing disorders.

2. Single pendulum

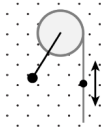
The damped driven pendulum is a paradigm of nonlinear dynamics. The interplay between the applied torque and the pendulum

Table 1

Fixed parameters for the Eq. (1) nonlinear pendulum and the Eq. (3) pendulum array.

Parameter	γ	τ_0	τ_1	ω	$\bar{l}$
Value	3/4	0.7155	2/5	1/4	1

gravitational torque determines whether the pendulum rotates or merely librates, potentially causing chaos. One instantiation of these dynamics is a simple pendulum rigidly connected to an axle wrapped by a rope that hangs down one side with a mass climbing up and down it, illustrated by the figure inset to the equation of motion



$$ml^2\ddot{\theta} = -\gamma\dot{\theta} - mgl\sin\theta + \tau_0 + \tau_1\sin\omega t, \quad (1)$$

which governs the angle θ of a mass m swinging a distance l from a pivot with damping γ , constant torque τ_0 , sinusoidal torque of amplitude τ_1 , and angular frequency ω . The three-dimensional phase space $(\theta, \dot{\theta}, \varphi)$, where driving phase $\varphi = \omega t$, contracts exponentially with Lyapunov exponents $\lambda_1 + \lambda_2 + \lambda_3 = -\gamma/ml^2$, where $\lambda_3 = 0$.

Fig. 1 summarizes the dynamics in plots of number N_r of rotations per driving period versus length l and period $T = 2\pi/\omega$ starting from zero initial conditions after a long transient. Colors in the top pair of figures illustrate Arnold tongues [26] of periodic motion separated by fractal-like chaotic bifurcations. The black dot and white circle indicate the original [14,15] and our default operating conditions, using the Table 1 parameters, with the constant torque τ_0 adjusted so the motion is on the border between libration and single rotation per torquing period. (Any operating point on a Fig. 1 bifurcation should also work.) Other parameters nondimensionalize the plot axes: $\hat{\tau}_g = mgl$ and $\hat{\tau}_d = \tau_0 + \tau_1$ are the maximum gravitational and driving torques, and $\langle\omega_0\rangle = \tau_0/\gamma$ is the mean angular velocity of the pendulum in the exactly solvable, nonchaotic limit of zero length $l \rightarrow 0$. Maximal Lyapunov exponent $\hat{\lambda} = \max\{\lambda_i\}$ is positive between periodic bands where dynamics is chaotic.

3. Pendulum array

3.1. Equations of motion

We generalize the pendulum Eq. (1) to a torsionally coupled array of N damped driven pendula governed by the equations of motion

$$ml_n^2\ddot{\theta}_n = -\gamma\dot{\theta}_n - mgl_n\sin\theta_n + \tau_0 + \tau_1\sin\omega t + \kappa(\theta_{n-1} - \theta_n) + \kappa(\theta_{n+1} - \theta_n), \quad (2)$$

for $n = 1, 2, 3, \dots, N$, where the torsion κ represents the strength of the linear nearest-neighbor coupling, the stiffness of torsional springs connecting adjacent pendula. The ends are free, with $\theta_0 = \theta_1$ and $\theta_{N+1} = \theta_N$.

By absorbing gravity g into time t and mass m into the other parameters, but retaining l_n (which will be disordered), the scaled equations of motion become

$$l_n^2\ddot{\theta}_n = -\gamma\dot{\theta}_n - l_n\sin\theta_n + \tau_0 + \tau_1\sin\omega t + \kappa(\theta_{n-1} - \theta_n) + \kappa(\theta_{n+1} - \theta_n). \quad (3)$$

3.2. Disorder

We introduce spatial disorder Δ by varying pendulum lengths l_n . Mean pendulum length $\langle l_n \rangle = \bar{l}$ is held artificially constant to isolate

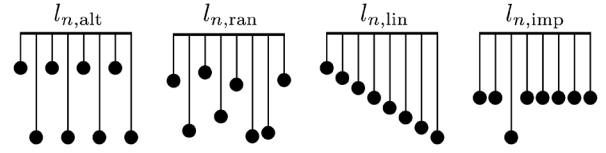


Fig. 2. Iconic illustrations of the pendulum arrays for the Eq. (4) disorder types with $\Delta = 0.5$.

Table 2

Random disorder.

n	$(l_{n,\text{ran}} - 1)/\Delta = u - \bar{u}$
1	-0.6787219230526584
2	+0.7621351560306098
3	-0.9171786598573886
5	+0.3520978154595248
4	-0.5674278557198684
6	+0.9214761206455304
7	+0.8244677200486106
8	-0.6968483735543598

the effects of disordering pendula lengths from those of varying mean length. Choosing $\langle l_n \rangle = 1$ places the array on a chaotic bifurcation in parameter space, as illustrated in Fig. 1 for the single pendulum, corresponding to the homogeneous, uncoupled array. Appendix B illustrates pendulum array behavior when mean length is not fixed.

We introduce four types of disorder to the pendulum array. These include the three used by Lindner et al. [15]—alternating binary, random, and linear—as well as the single impurity used by Gavrielides et al. [20] and Weiss et al. [21]. Unit-mean pendulum lengths are

$$l_{n,\text{alt}} = 1 + \left((-1)^n + \frac{N \% 2}{N} \right) \Delta, \quad (4a)$$

$$l_{n,\text{ran}} = 1 + (u - \bar{u})\Delta, \quad (4b)$$

$$l_{n,\text{lin}} = 1 + \left(\frac{2n - N - 1}{N - 1} \right) \Delta, \quad (4c)$$

$$l_{n,\text{imp}} = 1 + \left(\frac{\delta_{m,n}N - 1}{N - 1} \right) \Delta, \quad (4d)$$

where Δ quantifies the disorder, $\delta_{m,n}$ is the Kronecker delta, $\%$ is the modulo operator, $u \sim U(-1, 1)$ are random real numbers uniformly distributed between -1 and 1 with mean $\bar{u}$, and subscripts “alt”, “ran”, “lin”, and “imp” refer to alternating binary, random, linear, and single impurity disorder types, respectively.

We present results for the generic case of $N = 8$ coupled pendula, including an impurity in the $m = 3$ rd one and the Table 2 random disorder. Instantiations of the pendulum array are illustrated by Fig. 2.

3.3. Large coupling limit

We check our results by comparing high coupling κ to infinite coupling, where the pendulum array is fused into a single pendulum and the N motion Eqs. (3) reduce simply to

$$\bar{l}^2\ddot{\theta} = -\gamma\dot{\theta} - \bar{l}\sin\theta + \tau_0 + \tau_1\sin\omega t, \quad (5)$$

where

$$\bar{l} = \langle l_n \rangle = \frac{1}{N} \sum_{n=1}^N l_n \quad (6)$$

is the mean length,

$$\bar{l}^2 = \langle l_n^2 \rangle = \frac{1}{N} \sum_{n=1}^N l_n^2 \quad (7)$$

is the mean-square length, and

$$\sigma = \sqrt{\langle l_n^2 \rangle - \langle l_n \rangle^2} \quad (8)$$

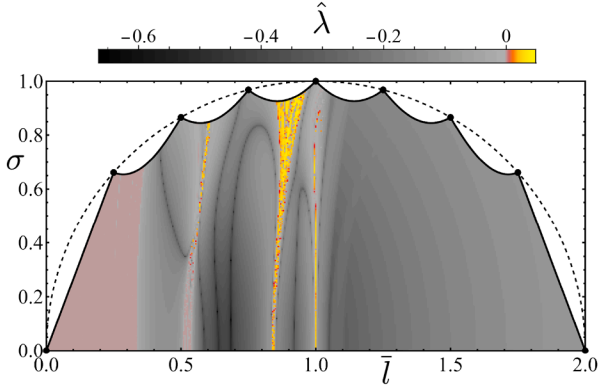


Fig. 3. Length standard deviation σ versus mean length $\bar{l}$ possibility space for $N = 8$ disordered pendula fused by infinite coupling $\kappa \rightarrow \infty$ with lengths $l_n \in [1 - \Delta, 1 + \Delta] = [S, L]$ and disorder $\Delta = 1$. Black dots where all $l_n \in [S, L]$ punctuate the boundary of the Eq. (10) semicircle. Magma palette classifies those maximal Lyapunov exponents $\hat{\lambda}$ accessible to such pendulum arrays.

is the length population standard deviation. For each disorder type, we derive a proportionality between standard deviation σ and disorder Δ by substituting Eq. (4) into Eq. (8) with $N = 8$ and $m = 3$ to find

$$\sigma_{\text{alt}} = \Delta, \quad (9a)$$

$$\sigma_{\text{ran}} \approx 0.74\Delta, \quad (9b)$$

$$\sigma_{\text{lin}} = \sqrt{\frac{3}{7}}\Delta \approx 0.65\Delta, \quad (9c)$$

$$\sigma_{\text{imp}} = \sqrt{\frac{1}{7}}\Delta \approx 0.38\Delta. \quad (9d)$$

For fixed dynamical parameters, if large coupling fuses the pendula together, the equations of motion depend only on the mean and mean-squared lengths, $\langle l_n \rangle$ and $\langle l_n^2 \rangle$. Fig. 3 presents the possibility space $(\bar{l}, \sigma)$ for 8 disordered pendula in the limit of infinite coupling, for disorder $\Delta = 1$ and lengths $1 - \Delta \leq l_n \leq 1 + \Delta$. Boundary vertices are combinations of maximally short $S = 1 - \Delta$ and long $L = 1 + \Delta$ pendula, which lie on the $\sigma > 0$ semicircle

$$(\bar{l} - 1)^2 + \sigma^2 = \Delta^2. \quad (10)$$

Magma palette indicates accessible maximal Lyapunov exponents, a mix of much order (grays) and some weak and strong chaos (reds and yellows, respectively). The choice of $\langle l_n \rangle = 1$ places the pendulum array on a thin chaotic island, from which it can be disordered into a vast sea of stable, periodic evolution. Increasing array length N increases the number of boundary vertices until, in the $N \rightarrow \infty$ limit, the entire Eq. (10) semicircle represents pendulum arrays.

4. Lyapunov exponents

In principle, modern mathematical techniques such as Morales-Ramis theory [27,28] can be used to prove the integrability or nonintegrability of equations of motion. However, an exhaustive integrability analysis of pendulum systems with many degrees of freedom is difficult due to the complicated analysis of the corresponding high-dimensional variational equations, and we do not pursue that project here.

Although order and chaos of Hamiltonian systems can be efficiently determined by numerically computing small alignment indices [29] or even estimated by machine or deep learning techniques [30,31], for best results with our dissipative system, we approximate the maximal Lyapunov exponent by integrating the variational equations [32]. Specifically, we integrate the system vector equation

$$\dot{x} = f(x) \quad (11)$$

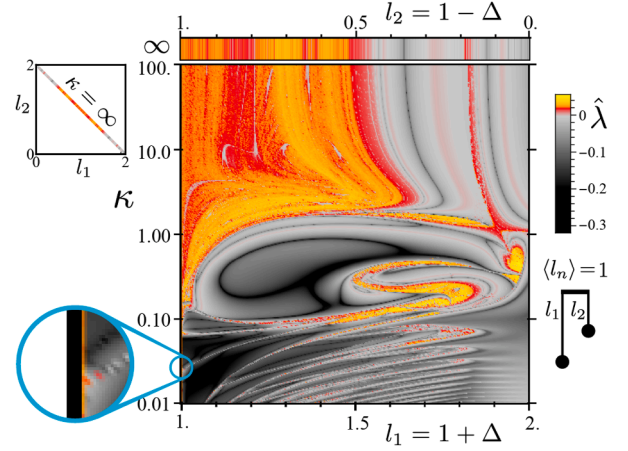


Fig. 4. Maximal Lyapunov exponent $\hat{\lambda}$ (magma colors) for all possible disorderings of lengths l_1 and l_2 with unit mean $\bar{l} = 1$ for increasing couplings κ for a system of $N = 2$ pendula. Note the chaos at the homogeneous $\Delta = 0$ left edge. At small coupling, very small disorder can tame chaos, but at large coupling, large disorder is needed.

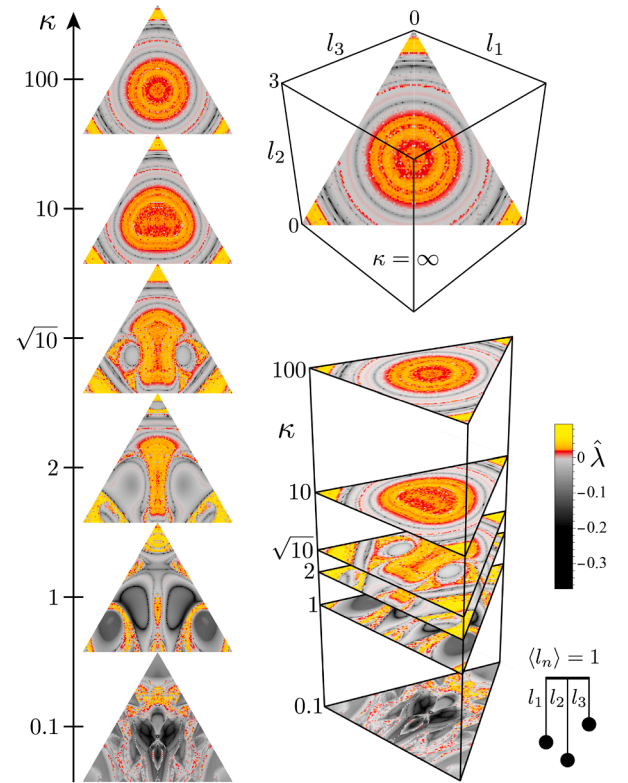


Fig. 5. Maximal Lyapunov exponent $\hat{\lambda}$ (magma colors) for all possible disorderings of lengths l_1, l_2, l_3 with unit mean $\bar{l} = 1$ for increasing couplings κ for a system of $N = 3$ pendula. Small coupling facilitates order (grays) and DTC, while large coupling favors chaos (reds and yellows).

concurrently with the variational matrix equation

$$\dot{M} = JM, \quad (12)$$

where

$$J = \frac{\partial f}{\partial x} \quad (13)$$

is the Jacobian matrix of partial derivatives with initial conditions $x(0) = x_0$ and $M(0) = I$, the identity matrix. For a system with dimension D , we numerically integrate D system differential equations concurrently

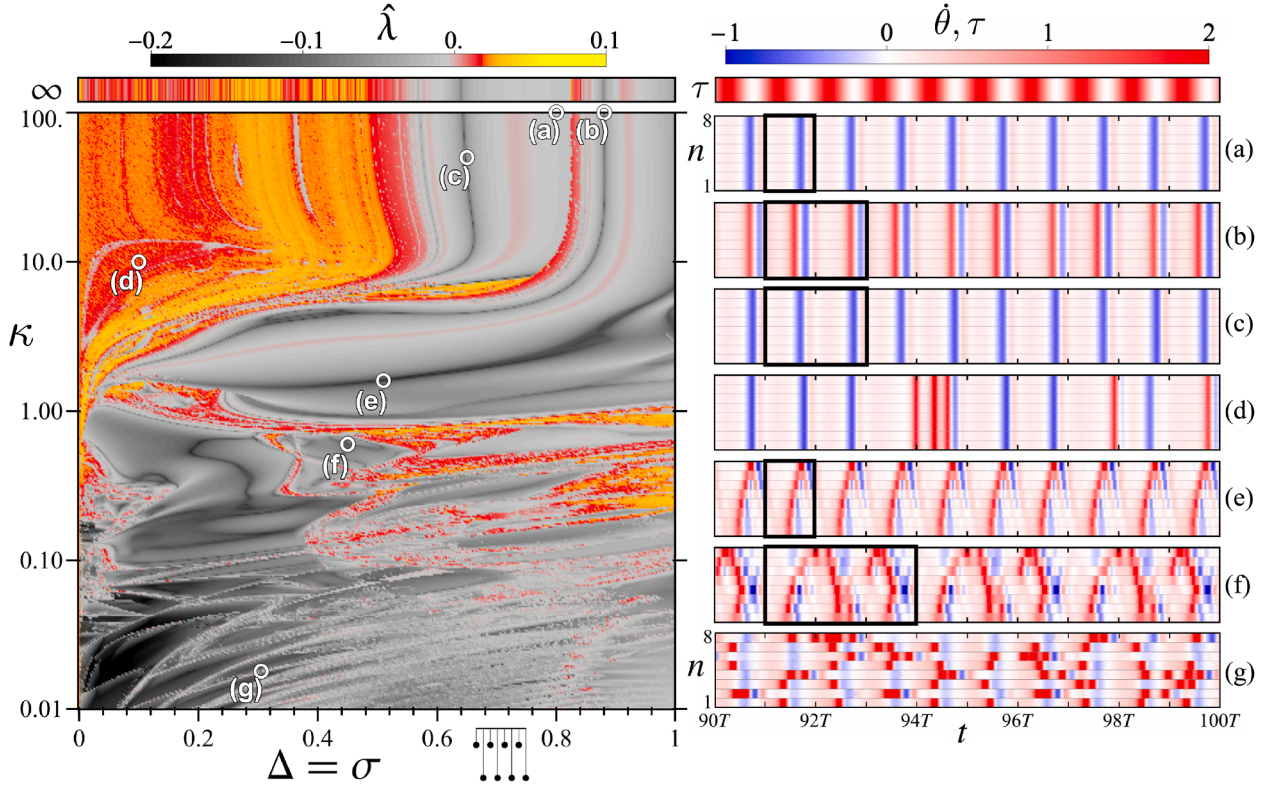


Fig. 6. Left: Maximal Lyapunov exponent $\hat{\lambda}$ (magma colors) versus coupling κ and disorder $\Delta = \sigma$ for a system of $N = 8$ nonlinear pendula under alternating binary disorder. The disorder instantiation is illustrated by the bottom graphic for $\Delta = 0.5$. As in Fig. 1, chaotic islands, isthmuses, and peninsulas (reds and yellows) bifurcate seas of different periodicities (grays). Right: Angular velocity $\dot{\theta}$ (Doppler colors) versus pendulum n and time t measured in torquing periods $T = 2\pi/\omega$ for six (σ, κ) : (a)(0.88, 100), (b)(0.88, 100), (c)(0.65, 50), (d)(0.1, 10), (e)(0.51, 1.6), (f)(0.45, 0.6), (g)(0.305, 0.018). Asymmetric applied torque τ (Doppler colors) drives the array for 100 torquing periods and the final 10 are recorded here. Black boxes enclose one period of evolution for periodic points. Pendulum arrays (b) and (e) have period T , (a) and (c) have period $2T$, (f) has period $3T$, while (d) and (g) appear to have infinite periods consistent with positive Lyapunov exponents and chaos.

with D^2 variational equations using the adaptive step-size Cash-Karp Runge-Kutta (4,5) algorithm [33].

We sequentially integrate these $D + D^2$ equations $\mathcal{N}$ times for time $\mathcal{T}$ to find

$$\delta x_{n+1} = M_{\mathcal{T}}(x_n)u_n, \quad (14)$$

where the (re)normalized unit perturbation $u_n = \delta x_n / |\delta x_n|$ is a D -dimensional vector. We expect the perturbation to grow exponentially on average like

$$\prod_{n=0}^{\mathcal{N}-1} |\delta x_n| = |\delta x| \sim e^{\hat{\lambda} \mathcal{N} \mathcal{T}}, \quad (15)$$

and so the dominant, maximal Lyapunov exponent

$$\hat{\lambda} \sim \frac{1}{\mathcal{N} \mathcal{T}} \log |\delta x| = \frac{1}{\mathcal{N} \mathcal{T}} \sum_{n=0}^{\mathcal{N}} \log |\delta x_n|. \quad (16)$$

Appendix A discusses details about our numerical implementation, including choices of divergence time $\mathcal{T}$ and number of (re)normalizations $\mathcal{N}$ for which Eq. (15) holds.

5. Results

5.1. Short arrays

Disordered short arrays can be exhaustively studied. The maximal Lyapunov exponent $\hat{\lambda}$ versus coupling κ for all mean-preserving disorders of arrays of $N = 2$ pendula is a diagonal line $(l_1 + l_2)/2 = 1$ in (l_1, l_2) lengths space—one for each coupling κ . Stacking these lines vertically

produces Fig. 4, where $l_1 = 1 + \Delta$ and $l_2 = 1 - \Delta$. Agreement with the fused $\kappa = \infty$ results is excellent. For small couplings $\kappa < 0.1$, even very small disorders $\Delta \ll 1$ convert chaotic dynamics to ordered dynamics. A large sea of periodic orbits near $\kappa \sim 0.1$ is a generic feature of such plots, even for longer arrays of different disorders, as described below. For large coupling $\kappa > 1$, large disorder $\Delta > 0.6$ is needed to control the chaos.

Fig. 5 presents the maximal Lyapunov exponent $\hat{\lambda}$ versus coupling κ for all mean-preserving disorders of arrays of $N = 3$ pendula, which is a diagonal plane $(l_1 + l_2 + l_3)/3 = 1$ centered on the homogeneous array in (l_1, l_2, l_3) space—one for each coupling κ , spread (left) and stacked (right). Perturbation size needed to order the dynamics increases with coupling. Bilateral symmetry becomes circular symmetry as coupling approaches infinity. As a check, the array $\kappa = 100$ plot is nearly identical to the fused $\kappa = \infty$ plot.

5.2. Generic arrays

Here we study a generic array of $N = 8$ pendula (but obtain similar results for shorter or longer arrays). For the Eq. (4a) alternate binary disorder, Fig. 6 plots maximal Lyapunov exponent $\hat{\lambda}$ versus disorder Δ and coupling κ for the pendulum array (left) and selected spatiotemporal evolutions $\dot{\theta}_n(t)$ (right).

Magma colors code the Lyapunov exponents, with gray indicating periodic $\hat{\lambda} < 0$ dynamics and red and yellow indicating chaotic dynamics $\hat{\lambda} > 0$. Disorder Δ and population standard deviation σ increase rightward on the horizontal axes and log-scaled coupling κ increases upward on the vertical axis. Small coupling $\kappa \ll 1$ favors periodicity, as disorder lengthens or shortens individual pendula causing them to librate or

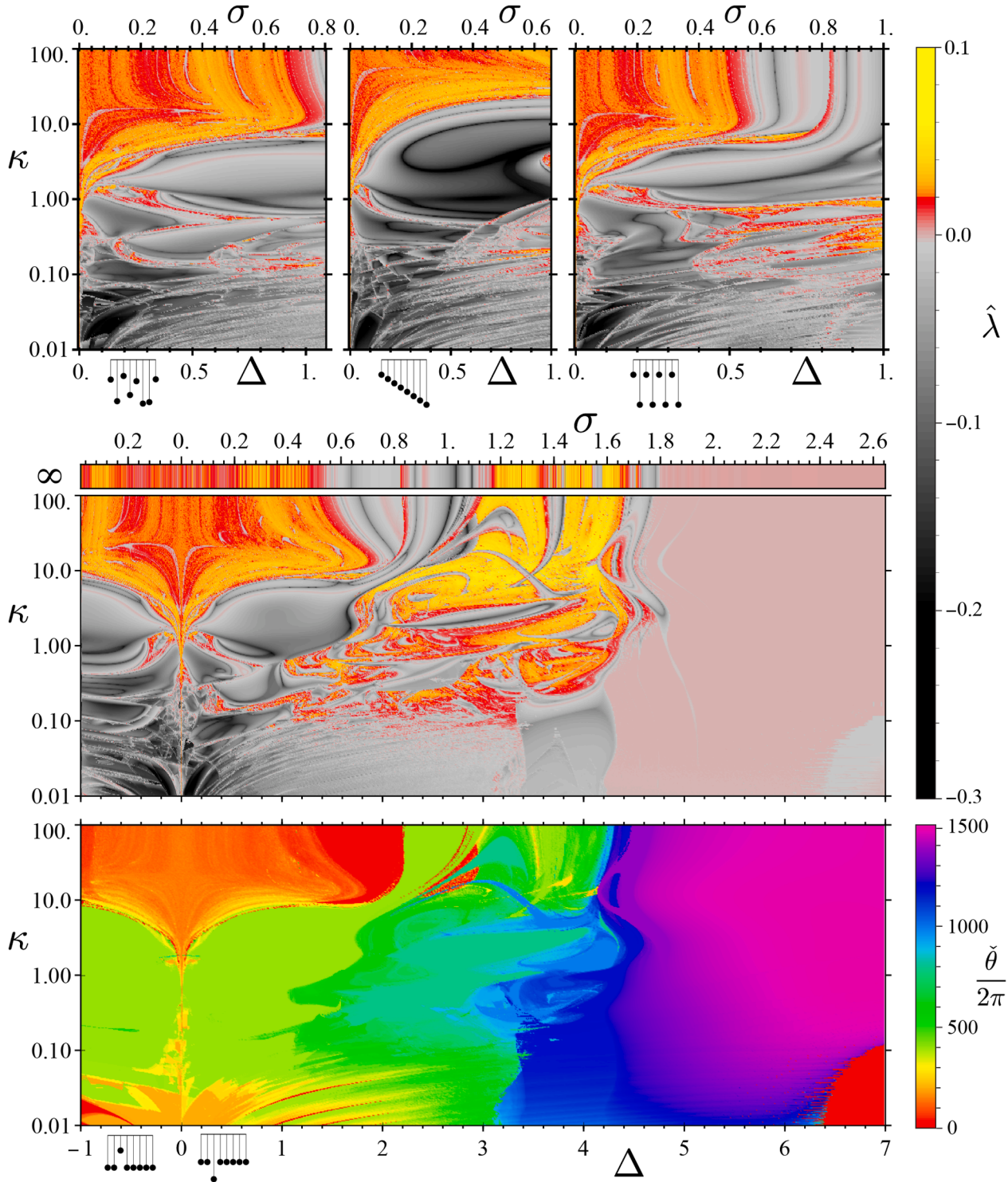


Fig. 7. Top row: Maximal Lyapunov exponent $\hat{\lambda}$ (magma colors) versus coupling κ and disorder Δ for a system of $N = 8$ pendula with random, linear, and alternating disorder. Middle row: Maximal Lyapunov exponent for single impurity disorder, including the infinite coupling limit. Bottom row: Minimum pendulum rotation angle $\tilde{\theta}$ (rainbow colors) for single impurity disorder after 400 torquing periods elucidates the corresponding Lyapunov landscape.

rotate periodically. Large coupling $\kappa \gg 1$ enforces homogeneity and the mean chaoticity, but sufficiently large disorder tames it.

Above the Lyapunov plot is a 2-D slice of Lyapunov exponents in the $\kappa \rightarrow \infty$ limit, modeled by Eq. (5) and obtained by implementing the Section 4 variational algorithm with Mathematica's Wolfram Language. Similarities between this limit and the finite κ Lyapunov exponents, obtained by implementing the variational algorithm with

the C programming language, suggest both array homogenization and agreement between the Wolfram- and C-language codes. Since array dynamics in the infinite coupling limit depend only on the mean and mean-squared lengths, this limit is common to all arrays, regardless of disorder type and assuming proper scaling in σ .

Doppler colors code angular velocities $\dot{\theta}_n = \partial\theta_n/\partial t$, with blue negative, white zero, and red positive. Red predominates due to the constant

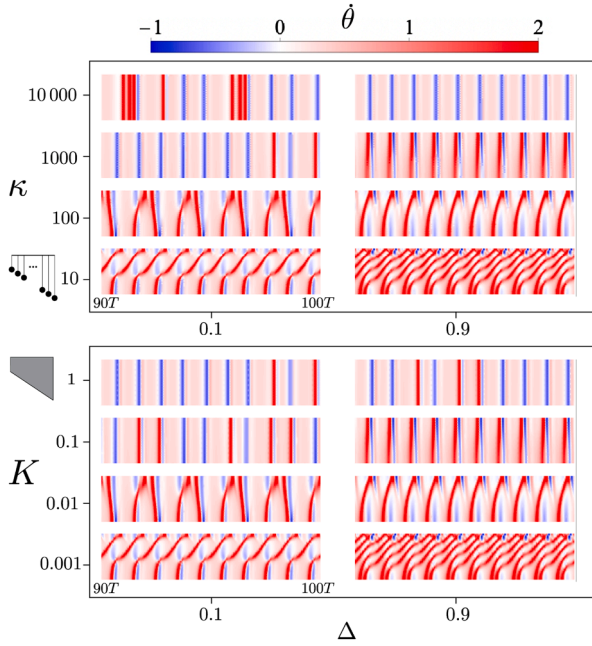


Fig. 8. Angular velocity $\dot{\theta}_n(t)$ spatiotemporal patterns (Doppler colors) for $N = 100$ pendulum array (top) and corresponding continuum limit (bottom) for different linear disorders Δ and couplings κ illustrating equivalent dynamics for the $\kappa = K N^2$ scaling.

torque component τ_0 . Large coupling homogenizes the array with near-zero twist $\delta\theta_n = \delta\theta_n/\delta n$, but small coupling encourages large twist until the corresponding torque $\kappa \delta\theta_n$ is comparable to the gravity and external torques.

Black boxes enclose one period of evolution for periodic points. Despite similar parameter sets, nearby points that lie in periodic seas in (σ, κ) space exhibit different evolution characteristics when separated by an island, isthmus, or peninsula of chaotic evolution. Points (a) and (b), for example, have the same value of κ and only a 0.9% difference in σ , but there is a chaotic island separating the two. In turn, the system at point (a) undergoes period-1 libration and the system at point (b) undergoes period-2 rotation. Nonlinear systems may be designed with this sensitivity in mind to either avoid regions close to chaotic islands and thus guarantee stable evolution, or to prefer dynamically rich seas so that various evolution patterns are achievable with minimal changes to parameters.

Fig. 7 compares the dynamics of pendulum arrays under the disorder types given in Eq. (4). Top row presents maximal Lyapunov exponent $\hat{\lambda}$ (magma colors) versus coupling κ and disorder Δ for a system of $N = 8$ pendula with random, linear, and alternating disorder. The upper axes indicate the corresponding pendulum-length standard deviations σ , all on the same scale. Pendulum dynamics are quite similar in (σ, κ) space, regardless of disorder type, with DTC at large disorders, inhibited by large coupling and promoted by low coupling, with periodic back-and-forth waves dominating the intermediate couplings. The random pendulum lengths closely resemble the alternating binary pendulum lengths and the dynamics of both systems align most closely. When these same random lengths are reordered to resemble the system under linear disorder, its dynamics resemble that of the linear disorder case.

The **Fig. 7** middle row presents maximal Lyapunov exponents for a single impurity, both short $\Delta < 0$ and long $\Delta > 0$. Agreement between the $\kappa = 100$ exponents and the fused $\kappa = \infty$ exponents is strong. The largest Lyapunov exponents occur for intermediate disorders.

To elucidate the single-impurity dynamical variety, the **Fig. 7** bottom row presents the minimum pendulum rotation angle $\hat{\theta}$ (rainbow colors), averaged over the array and the final torquing period, after 400 torquing

periods. In the top right, for large disorder and large coupling, the applied torques wind the shorter pendula many times against the long pendulum, until eventually the torque on the latter is sufficient to rotate it and the array end over end for all time (magenta). This “streaming” motion is characterized by a near-zero maximal Lyapunov exponent. By contrast, on the bottom right, for large disorder and *small* coupling, even maximal winding is insufficient to rotate the long pendulum, which librates through small angles for all time (red).

In all four $\hat{\lambda}$ plots, disorder ranges from the homogeneous array $\Delta = 0$ to the mathematical, but non-physical, zero-pendulum-length limit $l_n > 0$. This limit corresponds to maximum disorders $\Delta \in \{1, 1.09, \dots, 1, 7\}$ for alternate, random, linear, and single impurity disorders, respectively. The negative disorder limit $\Delta = -1$ was also explored in the latter’s case. In computation, pendulum length was bounded below by $l_n \geq 0.0025$. The zero-length limit was explored for the single pendulum in **Section 2** and is consistent with other results.

5.3. Infinite arrays

To study *infinite*-length arrays of pendula, in the continuum limit, interpret the linear nearest-neighbor coupling as a second-order finite difference with step $dx = 1$. Replace $\theta_n(t) \rightarrow \theta(x, t)$ and $l_n \rightarrow l(x)$ in the **Eq. (3)** coupled ordinary differential equation, and take the limit as $N \rightarrow \infty$ and $dx = 1/N \rightarrow 0$ such that $\kappa dx^2 = K$ is constant, to generate the single partial differential equation

$$\begin{aligned} l(x)^2 \partial_t^2 \theta(x, t) = & -\gamma \partial_x \theta(x, t) \\ & -l(x) \sin \theta(x, t) \\ & + \tau_0 + \tau_1 \sin \omega t \\ & + K \partial_x^2 \theta(x, t), \end{aligned} \quad (17)$$

which is a generalization of the sine-Gordon equation [34]. This equation can model a torqued, damped, and twisting elastic ribbon. Assuming linear disorder

$$l(x) = 1 + (2x - 1)\Delta, \quad (18)$$

minimum energy start ($\theta = 0$ and $\partial_t \theta = 0$ at $t = 0$), and free ends ($\partial_x \theta = 0$ at $x = 0$ and $x = 1$), numerical integration of **Eq. (17)** in Mathematica using NDSolve verifies a $\kappa = K N^2$ scaling with spatiotemporal features similar to that of the finite arrays, as in **Fig. 8**.

6. Discussion

We expect disorder-taming-chaos (DTC) to occur generically in coupled nonlinear systems when the components operate on the boundary between two qualitatively different dynamics. In practice, one can fix all but one of the system parameters and vary the remaining one until a bifurcation is reached, as we did here with the constant torque τ_0 . Such a system can be used as a sensitive detector of manufacturing errors, as a single imperfect component will qualitatively change the dynamics from chaotic to periodic.

We have analyzed the dynamics of a single pendulum, two and three coupled pendula, the generic case of eight coupled pendula, and the continuum limit of infinitely many coupled pendula. We have introduced a large-coupling possibility space to organize and classify all length disorderings of the pendulum arrays. We have identified a plateau in parameter space where Lyapunov exponents make sense, and thereby classified the dynamics as periodic or chaotic as a function of disorder and coupling, consistent with the observed spatiotemporal behavior. We have elucidated the role of coupling in DTC, where small coupling is a facilitator and large coupling is an inhibitor, for mean-preserving disorders.

Disordering the lengths of torqued and damped coupled pendula can convert chaotic dynamics to periodic dynamics, even if the mean length is fixed. Time-varying the disorder even slightly can also affect the dynamics, and such dynamical disorder connects with the vast control-of-chaos literature [35–37].

To further explore the coupled-pendula Lyapunov landscape, future work might include computing the Lyapunov *spectrum* as a function of disorder and coupling, while generalizing from nearest-neighbor coupling to small-world coupling or random coupling [9]. Spectra should enable classification among periodic, chaotic, or *hyperchaotic* motion, where hyperchaos [8,10] requires at least *two* positive Lyapunov exponents.

Our Lyapunov-landscape methodology could also be applied to multi-pendulum systems with variable geometry, including the Swing-ing Atwood Machine and its generalizations. These models also exhibit rich transitions between regularity and chaos and have been the subject of recent work [6,7,10]. A limitation of our approach is the very large variational state space associated, especially, with nonlinear elements coupled in two and three dimensions, as in a spatial lattice.

CRedit authorship contribution statement

Mikael Gallego: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **John F. Lindner:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Anil Radhakrishnan:** Writing – review & editing, Validation, Supervision, Software, Investigation, Formal analysis; **William L. Ditto:** Writing – review & editing, Resources, Project administration, Conceptualization.

Data availability

<https://github.com/TradeMark17/Pendula>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Numerical implementation

We modeled our Eq. (3) pendulum system in the C and Wolfram programming languages. Numerical integration in C was carried out by the adaptive step-size Cash-Karp Runge-Kutta (4,5) algorithm [33] expressly designed for initial value problems with rapidly varying right-hand sides, which is especially helpful when high disorder produces very short pendula that rotate quickly. We used a local error

$$\epsilon \leq 5 \times 10^{-7} \quad (\text{A.1})$$

and a variable time step

$$10^{-9} \leq dt \leq 10^{-2} \quad (\text{A.2})$$

or

$$10^{-10}T \lesssim dt \lesssim 10^{-2}T, \quad (\text{A.3})$$

where the torquing period $T = 2\pi/\omega = 8\pi$. Unsurprisingly, the time steps were smallest for large disorders where some of the pendula were shortest.

For our variational estimates of the maximal Lyapunov exponents, we used

$$\mathcal{N} = 500 \quad (\text{A.4})$$

(re)normalizations and a divergence time

$$\mathcal{T} = 50 \approx 20T. \quad (\text{A.5})$$

Numerical overflow was observed only at much higher divergence times $\mathcal{T} \gtrsim 3000 \approx 1200T$.

These parameters were confirmed to produce a broad plateau of maximal Lyapunov exponent convergence where the exponents were constant to within about 5%. (In comparison, uncertainty caused by the random variational perturbations was only about 0.1%.) This small uncertainty was built-into the magma palette where grays blur into reds. Maximal Lyapunov exponents matched previous pendulum array work [15] and agreed with prior results [38–41] for canonical 3-dimensional differential flows, including the autonomous Rössler system and non-autonomous Duffing system.

Due to the pendulums' damping and the many variational (re)normalizations, no initial transient was discarded before data capture. Because analysis performed with low-energy initial conditions was nearly identical to evolution with minimum-energy initial conditions, and to ensure that the dynamics remains in the same basin of attraction, all presented results were obtained with minimum-energy initial conditions $\theta_n(0) = \dot{\theta}_n(0) = 0$.

Results from the built-in NDSolve numerical integrator in Wolfram agree with those obtained using the aforementioned RK45 algorithm in C. C-language calculations were performed on North Carolina State University's Hazel high-performance computing cluster. The MPI package [42] was used to parallelize code. Mathematica [43] calculations were run locally. Plots were made in Mathematica. The UNIX random function was used to generate pseudo-random numbers in C. The Wolfram Language RandomReal function was used to generate pseudo-random numbers in Mathematica, including the Table 2 pendulum lengths.

Our C-language code and data are available on our GitHub page [44], along with a companion Mathematica notebook that checks our equations.

Appendix B. Mean-changing disorders

Throughout this article, we have focused on the “hard” and “rare” case of pendulum array disorders that preserve a constant mean length $\bar{l} = 1$, corresponding to an effective pendulum at a bifurcation between libration and single rotation, while the individual pendula would either librate or rotate if uncoupled. As seen in Fig. 3, this case lies on a thin chaotic island near the middle of $(\bar{l}, \sigma)$ space. Here we briefly consider the “easy” and “common” case of pendulum array disorders that vary the mean length $\bar{l} \neq 1$, corresponding to arrays in the wide periodic seas of $(\bar{l}, \sigma)$ space.

Fig. B.9 demonstrates the array's high sensitivity to mean length in the case of alternate binary disorder. When $\bar{l}$ is perturbed from this chaotic island by a relatively small amount, such as $\delta\bar{l} = \pm 0.01$, the dynamics become drastically more ordered in (σ, κ) space. Dynamics in the perturbed and unperturbed cases show strong agreement for low coupling and high disorder. At high coupling, the dynamics are nearly completely periodic for $\bar{l} = 1 \pm 0.01$. As in Fig. 7, Mathematica [43] was used to generate the corresponding $\kappa \rightarrow \infty$ limits described by Eq. 5. These results suggest that control over chaos is possible with very small perturbations in pendulum length, provided that the mean is not fixed. This analysis was repeated for a single impurity, which approached the $\kappa \rightarrow \infty$ homogenization limit much more quickly and evolved periodically at almost every point in (σ, κ) space, in agreement with previous work [20,21].

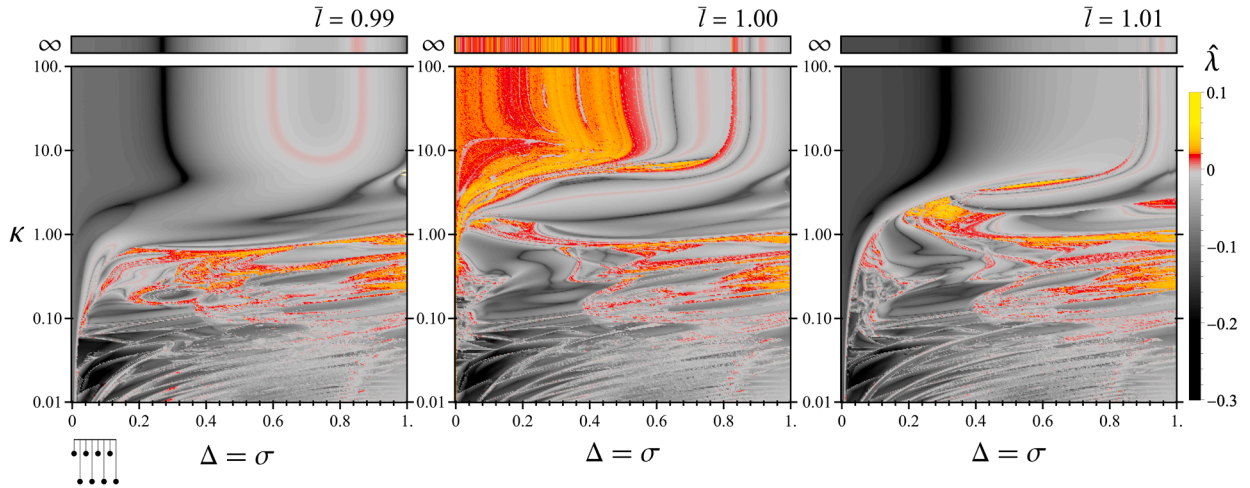


Fig. B.9. Maximal Lyapunov exponent $\hat{\lambda}$ (magma colors) plotted against coupling κ and disorder Δ for a system of $N = 8$ pendula under alternating binary disorder with mean lengths $\bar{l} \in \{0.99, 1, 1.01\}$. Pendulum icon illustrates the disordered lengths for $\Delta = 0.5$ and $\bar{l} = 1$. Note the chaos for $\bar{l} = 1.00$ at the homogeneous $\Delta = 0$ (left edge) for all coupling κ . Dynamics are similar for low coupling and high disorder, and vary more strongly otherwise. At high enough coupling $\kappa \gtrsim 5$, dynamics for $\bar{l} \neq 1$ are drastically more ordered. The plots show good agreement with the Eq. (5) theoretical infinite coupling limit.

Appendix C. Notation

Table C.3 summarizes our most common symbols and abbreviations.

Table C.3
Notation.

N	pendula number = array length
Δ	length disorder
σ	length standard deviation
l_n	length of pendulum n
$\bar{l}$	mean pendulum length
$\bar{l}^2$	mean-square pendulum length
κ	nearest-neighbor coupling
γ	viscosity
ω	alternating torque angular frequency
T	alternating torque period
τ_0	direct torque amplitude
τ_1	alternating torque amplitude
θ_n	angle rotation of pendulum n
$\dot{\theta}_n$	angular velocity of pendulum n
$\tilde{\theta}$	minimum pendulum rotation angle
t	evolution time
ϵ	RK45 local error
dt	RK45 variable integration time step
$\mathcal{N}$	number of (re)normalizations
$\mathcal{T}$	divergence time
$\hat{\lambda}$	maximal Lyapunov exponent
DTC	disorder-taming-chaos
RK45	Cash-Karp Runge-Kutta algorithm

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